

MediPilot: Detailed Business Proposal

Continuous Re-Triage for the Indian Emergency Department



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EXECUTIVE SUMMARY

Emergency department crowding kills through a structural defect rather than individual error: acuity is assigned once, at arrival, and the queue that results is static while the patient's physiology is not. **MediPilot is a continuous re-triage platform built on one architectural invariant — the system may raise a waiting patient's priority autonomously, and may never lower it below a human-assigned level.** Escalation costs one nurse assessment; de-escalation can cost a life, so it stays with a human.

This document is the Round II business proposal, and it is backed by a working prototype rather than a specification. Seven clinical surfaces run end to end over a 20-patient synthetic corpus that covers every case the brief mandates — an ambiguous epigastric presentation, a paediatric/geriatric pair at the same fever, a zero-history first attendance, a 3× surge, a clinician override with its audit record, and an out-of-distribution patient the system refuses to score. Six age strata are calibrated separately, because a single adult-calibrated model is a silent safety risk: our own corpus shows a 3-year-old at 38.7 °C with a heart rate of 138 that is *normal for age* and would read as severe tachycardia on an adult scale.

We report evaluation honestly, including where it is unflattering. A four-model Risk Engine bake-off on a freshly generated 20,000-patient held-out split shows the shipped gradient-boosted model leading on discrimination (AUPRC 0.164) and calibration (ECE 0.011) and being the only candidate to hold conformal coverage at or above 90%. It also shows that *no* learned model reliably beats the hand-coded rule card at tight operating points — which is precisely why the rule card remains the floor of the architecture rather than a fallback we hope never to use. On the perception side, a ten-candidate LLM structurizer bake-off gives gpt-oss-120b an F1 of 0.962 with zero red flags missed against 0.510 and twelve missed for the rule baseline, and a twelve-backend ASR bake-off selects the transcription path on measured word error rate rather than on a vendor claim.

The commercial wedge is deliberately narrow: ship the deterioration-while-waiting module alone. It carries the highest clinical value and the lowest regulatory exposure, because it never assigns an initial acuity — it only watches a queue a human already ordered.

Jurisdiction: India — DPDP Act 2023, ABDM consent framework, CDSCO SaMD pathway. **Status:** working prototype on synthetic data; no clinical claim is made.

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1 PROBLEM FRAMING

1.1 The defect is structural, not human

The danger of an ED waiting room is not that patients arrive critically ill. It is that they *become* critically ill after being assigned a low priority, and no mechanism exists to notice. A patient with mild chest discomfort receives a Yellow and sits down; twenty minutes later they are hemodynamically unstable; the triage nurse is by then registering three new arrivals.

This is the **single-point assessment model**, and its failure rate is a property of the process rather than of any clinician in it.

Figure	What it measures
32.2%	of US ED visits are mistriaged under ESI — roughly one nurse label in three
0.63	AUC of the Epic Sepsis Model in external validation over 38,455 hospitalisations, against a marketed 0.76–0.83
33%	its sensitivity: two of every three sepsis cases missed, while deployed at hundreds of US hospitals
109	alerts raised per true positive it did find — the alert-fatigue failure mode
11.7% / 3.6%	occult hypoxaemia, Black vs White patients, on pulse oximetry; 17% vs 6.2% in the multicentre cohort
100–500 ~50%	ED visits per day this must serve, per the brief of arrivals carry any prior record at all

Two of those numbers shape this entire proposal. The Epic Sepsis Model did not fail through carelessness; it failed because it learned *clinician suspicion* — antibiotic-order timing — rather than physiology, and so detected the very process it was meant to augment. That is a leakage failure, and §2 gives the mechanism that makes it structurally impossible here rather than merely discouraged. The oximetry figures are why a normal SpO₂ carries no de-escalation authority anywhere in this system.

India had no nationally standardised ED triage protocol before the AIIMS Triage Protocol — a three-tier Red/Yellow/Green system since adopted by AIIMS

Bhubaneswar, CMC Vellore, GTB Delhi and the Kerala DHS. Most Indian EDs are staffed by generalists rather than emergency-medicine specialists, so any workable system must run in a district hospital with one physician and one nurse as reliably as in a tertiary centre.

1.2 What the prototype already measures

Three numbers from our own instrumentation frame the rest of this document. On the 20-patient corpus, continuous five-minute re-scoring reaches the Yellow threshold a mean of **21.6 minutes** earlier than the next scheduled physical re-measurement would, and up to 55 minutes earlier in individual cases (Fig. 7). The risk engine itself decides in **3 ms** at p50, p99 8 ms (Fig. 9). And the chosen structurer misses **zero** red flags across a deliberately adversarial evaluation set, against twelve for the deterministic baseline beneath it (Fig. 8).

The contribution is not a better score. It is a better contract. A machine that is allowed to raise an alarm on its own, is never allowed to lower one, abstains out loud when it does not know, and is evaluated on the patients it misses rather than on the average it gets right.

1.3 The seven complexities, and our answer to each

The brief names seven real-world complexities. We treat them as design constraints rather than caveats, and each maps to a mechanism that exists in the prototype.

Complexity	Mechanism in the prototype
Ambiguous under-reported symptoms	Conformal abstention: the system returns an acuity <i>set</i> , and when the set is wide it declines and routes to a human (P-04, P-08). Reliability flags capture stoic presentation and communication barriers (P-03, P-16, P-18)
Age-varying thresholds	Six strata, each with its own normal ranges, its own isotonic calibration, its own conformal quantile and its own decision threshold (§4)
Uneven data at intake	Train-time modality dropout at 40%, so the model must produce a usable estimate from a snapshot; zero-history is a first-class path (P-07)
Seconds to decide, and explainable	One card: band, two strongest supporting factors, one opposing factor, and a confidence band. The opposing factor is a deliberate falsification target
Asymmetric costs	Cost ratio R enters training as an asymmetric loss and sets the threshold, so objective and operating point agree rather than being reconciled afterwards
Hospitals differ	Every clinical knob is YAML, signed off per site; arrival rate, cadences, red flags and R are site parameters (§10)
Accountability & liability	Override is a first-class route, not an escape hatch; every decision and reversal lands in a SHA-256 hash-chained ledger (§8)

2 SOLUTION DESIGN

2.1 One invariant, propagated through six layers

MediPilot is a six-layer pipeline closed by two feedback loops (Fig. 1). The asymmetry is not a policy setting anywhere in it — it is an architectural constraint. An attempt to lower a band without an attached human action raises *AsymmetricAutonomyViolation* on the code path.

Formally, escalation is optimal when $c_{FN} p \geq c_{FP}(1 - p)$,

giving the operating threshold

$$p^* = \frac{c_{FP}}{c_{FP} + c_{FN}} = \frac{1}{1 + R}, \quad R = \frac{c_{FN}}{c_{FP}}.$$

R is a **per-hospital configurable parameter** set by clinical leadership at deployment, not a global constant chosen by a vendor. The shipped artifact solves $R_{\text{yellow}} = 12.2$ and $R_{\text{red}} = 5.8$ against an under-triage budget, giving thresholds $p_{\text{yellow}}^* = 0.076$ and $p_{\text{red}}^* = 0.147$.

2.2 Data strategy — weighing inputs that are never complete

Inputs arrive in three windows with sharply different availability, and every field is registered with its earliest availability time and a leakage class before it may be used at all. `config/feature_registry.yaml` classifies each field **SAFE**, **CONDITIONAL** or **PROHIBITED** and **fails closed**: an unregistered name is treated as **PROHIBITED**, so adding a field without classifying it breaks the build. This is the mechanism that prevents the failure that broke the Epic Sepsis Model, which learned *clinician suspicion* (antibiotic order timing) rather than physiology and scored AUC 0.63 at 33% sensitivity in external validation.

Operational context — census, boarding count, staff on shift, bed availability — is ingested by the routing layer *only*, and is explicitly excluded from the risk layers. Physiologic risk is therefore never a function of how busy the department is.

The 59 features are six per vital: value, per-stratum z-score, reading age in minutes, slope per hour, 30-minute delta, and a count of readings. That trend block is what makes deterioration-while-waiting visible at all; a snapshot model cannot see a trajectory.

2.3 Decision model — hybrid, not a choice between rules and ML

The brief offers rules, ML, or a hybrid. We use a strict hybrid with a fixed precedence, and the deterministic layer wins:

#	Stage — and who may overrule whom
1	Stratum resolution from age
2	Vital freshness — stale readings are discounted at 2× cadence, treated as missing at 3×
3	Deterministic red flags fire first — eight observations, each mapping straight to Red, overridable only by a clinician
4	Vital thresholds, per stratum
5	Raw ML risk
6	Per-stratum isotonic calibration
7	Asymmetric reliability weighting
8	Conformal uncertainty
9	ScoreObject <i>or</i> AbstentionObject — never a partial object

Uncertainty is represented three ways, and all three are visible to the nurse. Isotonic calibration makes the number mean what it says at that site. Mondrian split-conformal prediction ($\alpha = 0.10$) returns an acuity *set* with a formal coverage guarantee held separately within each stratum. And an out-of-distribution gate routes unfamiliar patients to manual assessment with an explanatory note. **The interface enforces this:** *AcuityCard* throws if asked to render a score without a confidence band, so a bare number cannot reach a screen even by mistake.

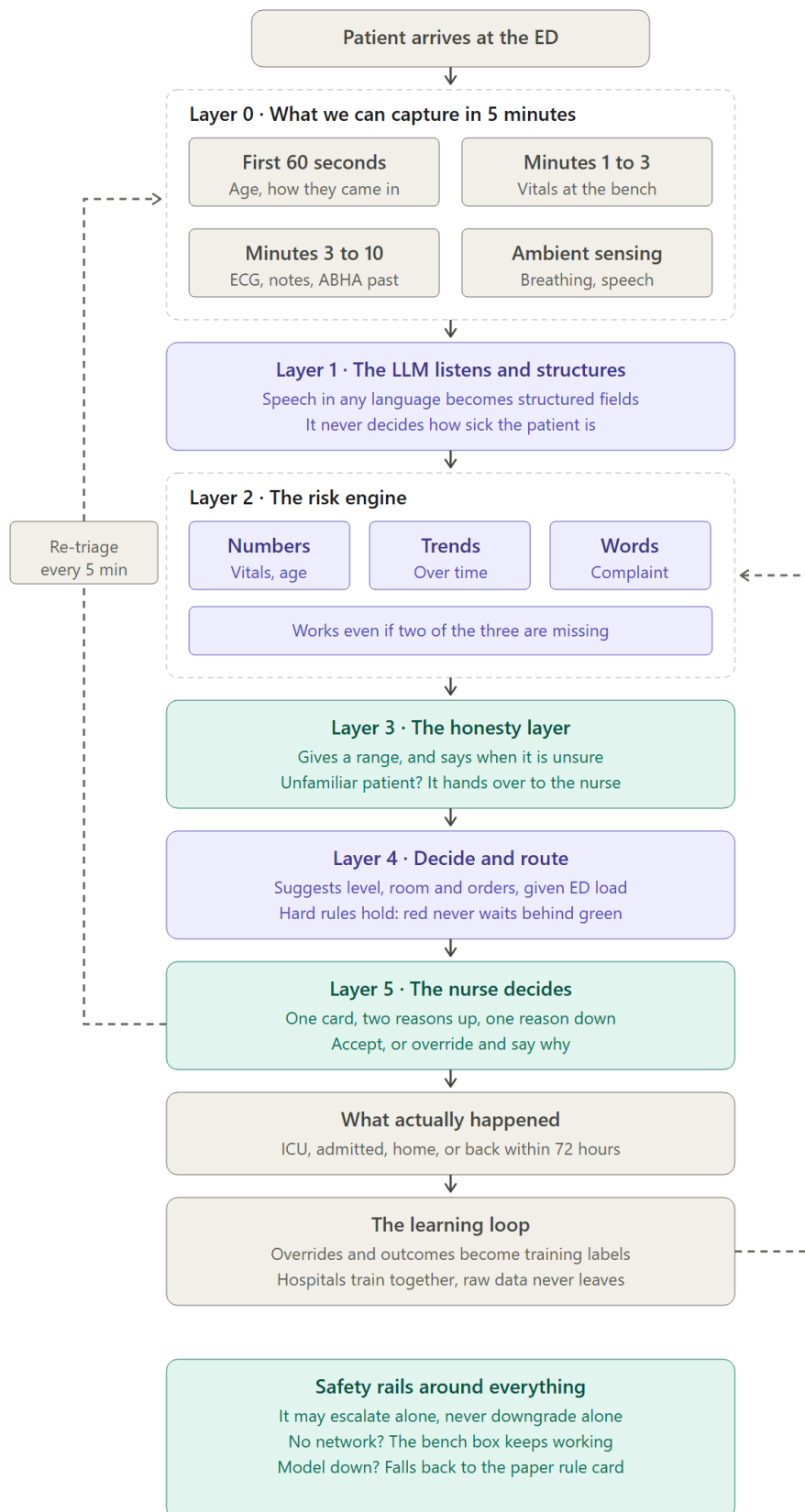


Figure 1. The six-layer pipeline and its two feedback loops. Layer 1 (the language model) may only structure speech — it never decides how sick the patient is. Layer 2 fuses numbers, trends and words and works even if two of the three are missing. Layer 3 is permitted to say it does not know. The dashed return on the left is the five-minute re-triage loop that makes the corridor observed; the dashed return on the right is the learning loop, in which overrides and outcomes become training labels.

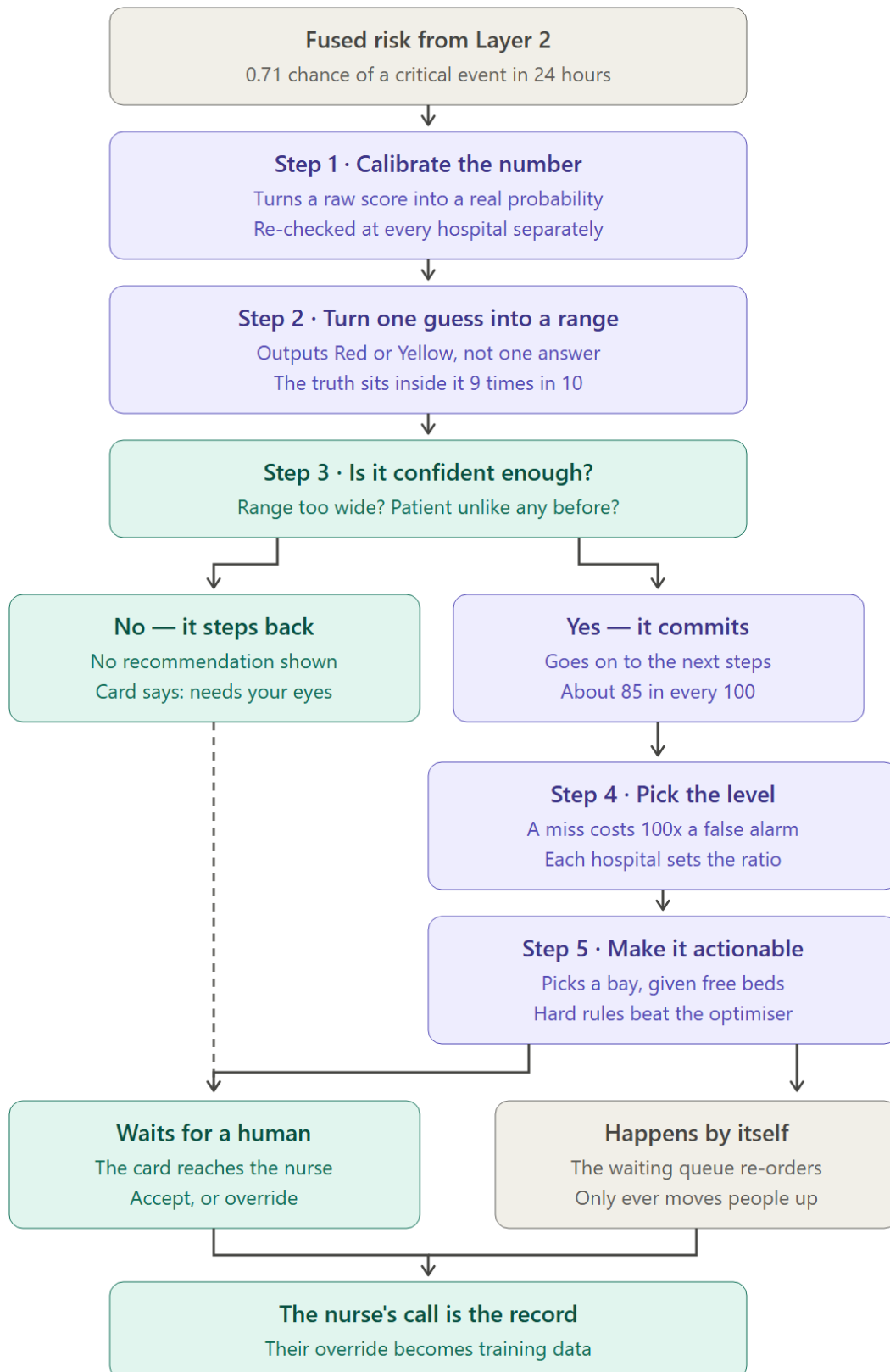


Figure 2. The decision path in depth, from a fused risk number to a band on the board. Read it as four gates in series. **Step 1** turns a raw score into a probability that means what it says, re-checked at every hospital separately — what §4 calls per-site calibration. **Step 2** replaces the point estimate with a range that contains the truth nine times in ten. **Step 3** is the gate most systems omit: an explicit confidence check that may *step back entirely*, showing no recommendation and routing the patient to a human. Only past that gate does the system pick a level — and the split at the bottom is the invariant made visible: the queue re-orders by itself but only ever moves people *up*, while anything that could lower a patient waits for a person. The nurse's call, not the model's, becomes the record, and therefore becomes the training data.

2.4 The decision path in depth

Fig. 2 expands stages 5–9 of the table above into the flow a single patient actually takes. Two properties are worth naming because they are unusual. First, **abstention is a terminal state, not an error** — there is a path through this system that ends in "no recommendation", drawn as a first-class outcome rather than an exception branch. Second, **the two bottom boxes carry different authority**: "happens by itself" is the autonomous escalation lane and is one-directional by construction, while "waits for a human" is everything else. No arrow in that diagram moves a patient down without passing through a person.

2.5 Workflow — the moment, the override, the surge

In the moment. The nurse speaks normally; the system fills the form and reads the vitals back so she can correct them before they commit. It never announces a level out loud where patients can hear, and the hall speaker calls token numbers only — never names or conditions.

The override. One touch from every surface showing a band. The dialog captures a reason code from a fixed list plus free text, because a fixed list never covers everything. The record fixes `inputs_hash` so the decision is reproducible and `factors_shown` so the card as displayed is preserved.

The surge. Detected at 3× the site's normal arrival rate over a 20-minute rolling window; exited below 2×. Yellow re-measurement stretches 30→45 minutes and Green 60→90. **Red never stretches.** Three actions are forbidden under any load and raise `SurgeViolation` rather than being discouraged: raising *R* to quieten alarms, resolving an abstention by guessing, and de-escalating anyone to free capacity. Alerts are ranked by risk × minutes waiting and are never suppressed.

2.6 Safety-first defaults and the waiting loop

Two clocks run per patient: re-score (model) and re-measure (physical). The brief requires monitoring of the waiting queue, and this is the module that does it.

Band	Re-score	Re-measure	Wait ceiling and breach action
Red	1 min	5 min	should not be waiting; pages a senior clinician at 5 min
Yellow	5 min	30 min	60 min → forced re-measurement; escalates to Red if not done in 15 min
Green	5 min	60 min	120 min → re-measure + wellbeing contact; two consecutive missed rechecks escalate to Yellow
Abstained	5 min	on human review only	15 min unreviewed → breach. Floor band is Yellow: an abstained patient is never Green

Recheck trust is tiered, which matters in an Indian ED where families participate in care. A recheck station or nurse can close any band. An accompanying family member can close a Green only, and doing so raises a flag for staff. Patient self-service can *escalate* but can never de-escalate and can never satisfy a recheck.

3 THE WORKING PROTOTYPE

3.1 Against the brief's minimum expectations

Required	Delivered
Scoring on 15–20 simulated records	20 deterministic records, P-01–P-20 , seed 42
≥ 1 ambiguous presentation	P-04 epigastric pain: gastritis vs inferior MI
≥ 1 paediatric or geriatric	7 across five non-adult strata; P-02/P-03 are the brief's own 38.5 °C pair
≥ 1 zero-history patient	P-07 , first attendance, no prior record
3× surge behaviour	P-10 under surge load; policy in <code>surge_policy.yaml</code>
No score without confidence	Enforced in the component, not by convention
≥ 1 clinician override, logged	P-09 , 19-field hash-chained record

3.2 The 20-patient corpus

Table 1 is the whole corpus. **P-01–P-10** map to the acceptance cases in the brief; **P-11–P-20** broaden stratum and condition coverage. Shaded rows are the cases the brief names explicitly.

3.3 Worked example — the brief's own 38.5 °C question

The brief asks what a fever of 38.5 °C means in a 3-year-old versus a 75-year-old, and warns that a single adult-calibrated model introduces silent safety risk. **P-02** and **P-03** are that question, made executable.

Vital	P-02 (3 y)	P-03 (75 y)	Adult-only reading
Temp	38.7	38.1	both febrile
HR	137.9 (normal 70–140)	67.6 (normal 60–100)	P-02 misread as severe tachycardia
RR	32.7 (normal 18–30)	22.9 (normal 12–20)	both deranged; P-03's easily missed
SpO ₂	95.8 (normal ≥95)	93.5 (normal ≥94)	P-03 hypoxaemic
BP sys	82.9 (normal 80–120)	143.6 (normal 90–150)	P-02 misread as hypotensive

On an adult scale the toddler generates two false alarms (HR, BP) that are simply normal for a three-year-old, while the 75-year-old's genuine hypoxaemia and tachypnoea sit inside adult-normal bands and pass unremarked. **The single-model error is bidirectional**: it over-triages children on rate and under-triages elders on the vitals that actually matter. That is the silent safety risk the brief describes, and stratified thresholds are the direct answer to it.

4 AGE AND SITE CALIBRATION

4.1 Six strata, calibrated independently

Stratum	Age range	HR normal	RR normal	Test FNR
Neonate	0–27 d	100–180	30–60	0.195
Infant	28–364 d	100–160	25–50	0.125
Child	1–11 y	70–140	18–30	0.077
Adolescent	12–17 y	60–110	14–22	0.000
Adult	18–64 y	60–100	12–20	0.152
Geriatric	65 y+	60–100	12–20	0.051

Table 1. The 20-patient demonstration corpus ([backend/data/corpus_20.json](#), seed 42, deterministic). Shaded rows are the presentations the brief mandates. Every record carries a full 37-step vital trajectory from T=0 to T=180 minutes at 5-minute intervals, so deterioration is replayable rather than asserted.

ID	Age	Stratum	Case	What it is there to prove
P-01	42 y	adult	deteriorates while waiting	The core claim. Yellow at arrival, autonomously escalated partway through the wait
P-02	3 y	child	paediatric fever pair	38.7 °C, HR 138, RR 33 — normal-for-age HR that an adult scale reads as severe tachycardia
P-03	75 y	geriatric	geriatric fever pair	38.1 °C, HR 68 — unremarkable rate, but RR 22.9 and SpO ₂ 93.5 are both out of range for age. Stoic presentation flagged Gastritis vs inferior MI. The ambiguous presentation the brief requires
P-04	55 y	adult	ambiguous epigastric pain	
P-05	35 y	adult	SpO ₂ device bias	Dark skin, SpO ₂ reads 96%, patient distressed. A normal reading carries <i>no</i> de-escalation authority
P-06	48 y	adult	stale vitals (3 h)	Freshness decay: beyond 3× cadence the reading is treated as missing, not as reassurance
P-07	38 y	adult	zero history	First attendance, nothing on file beyond what is observed now
P-08	50 y	adult	OOD abstention	Unusual vital combination. The system declines to score and says so
P-09	62 y	adult	nurse override	Yellow→Red on a physical finding the system could not see. Produces the audit record
P-10	28 y	adult	missed rechecks under surge	Green misses two consecutive rechecks during 3× load → escalates to Yellow
P-11	12 d	neonate	fever + floppy	Red flag RF-08. Neonatal vitals sit in an entirely different range (HR 100–180)
P-12	6 m	infant	sepsis	Infant stratum; the fastest-deteriorating trajectory shape
P-13	16 y	adolescent	poisoning	Red flag RF-07, with a communication barrier flag
P-14	58 y	adult	stroke	Red flag RF-05, plus difficulty speaking full sentences
P-15	29 y	adult	anaphylaxis	Red flag RF-04; airway compromise as a speech signal
P-16	80 y	geriatric	silent MI	Stoic presentation <i>and</i> analgesia already given — two reliability discounts at once
P-17	25 y	adult	trauma	Red flag RF-06, uncontrolled bleeding
P-18	72 y	geriatric	communication barrier	Communication barrier + health-literacy signal; self-report cannot be trusted as the only channel
P-19	17 y	adolescent	obstetric emergency	Red flag RF-02, active labour or bleeding in pregnancy
P-20	34 y	adult	green baseline	The control. A benign illness that must <i>stay</i> Green

Calibration is not one step but three, and each is fitted *per stratum*: isotonic regression so the probability means what it says, Mondrian conformal quantiles so the coverage guarantee holds inside each stratum rather than only on average, and separately solved decision thresholds. The two calibration splits are kept apart — fitting isotonic and then computing conformal quantiles on the same rows makes those scores in-sample and voids the guarantee while still printing coverage numbers that look fine.

4.2 Calibrating for a hospital

A site is not a code change. Deployment sets seven things, and in a real deployment each is clinically signed off:

Site parameter	Effect, and who owns it
Cost ratio R	Sets the escalation threshold. Tertiary centres run $R \approx 500$; a district hospital nearer 100. Clinical leadership
Normal arrival rate	Baseline for surge detection (default 8/hour). Operations
Band cadences	Re-score and re-measure intervals, wait ceilings. Nursing leadership
Red-flag table	Eight observations shipped; sites may add. Clinical governance
Age strata	Normal ranges per stratum, replaceable with site values
Per-site isotonic	Re-fitted on the site's own outcome data during shadow mode
Fitzpatrick calibration	SpO ₂ pallor correction, stratified locally

Shadow mode is the calibration mechanism. A new site runs MediPilot alongside its existing process with the model scoring silently and no recommendation shown. That produces the site's own outcome-linked data, which re-fits the isotonic calibrator and re-solves thresholds before anything is surfaced to a nurse. The eight go-live gates in §6 are the exit criteria.

5 DATA STRATEGY AND CURRENT LIMITATIONS

5.1 Labels come from outcomes, not from nurses

With ESI mistriage at 32.2%, roughly one nurse label in three is wrong; training a model to reproduce them would encode the error. MediPilot trains on a composite outcome label extractable from admission, discharge and billing records — so no clinician chart-review project is required, which removes the usual blocking dependency on scarce clinical time.

Three structural choices stop the synthetic label being circular: the label lives strictly in the future relative to the features; a *frailty* latent enters the outcome and never touches any vital, putting a hard floor under Bayes error; and the outcome is a Bernoulli draw rather than a threshold. A severity oracle that knows the true future peak severity is computed explicitly as the achievable ceiling — **AUPRC 0.408** — so model performance is reported against what is attainable rather than against perfection.

Stated plainly: there is no real patient data in this system. Every number in §6 was measured on a generator. The generator is physiologically parameterised and deliberately hard, but it is not an ED. No clinical, performance or causal claim is made, and **SIMULATED DATA** appears on every screen.

5.2 Current limitations, and what each blocks

Limitation	Consequence, and the unblock
No real ED data	Absolute metrics are not transferable. Unblocked only by shadow mode at a partner site
Stratum coverage is narrow	Five conditions define vitals for adults only. Rather than mint neonates with adult trauma physiology, sampling is restricted to 32 explicitly-defined (condition × stratum) pairs and every row records <code>stratum_is_fallback</code>
Small positive counts	Neonate FNR 0.195 has a CI of [0.075, 0.325] on 41 positives. The number is real; the precision is not
No ASR ground-truth corpus	The ASR bake-off runs on 2 reference utterances. It separates the field but does not settle it
Sequence models unscored	The last-observation/GRU/TCN track needs torch, which fails to initialise on the AMD development hardware
No EHR integration	FHIR/CDS Hooks is designed, not built (§10)

6 TESTING AND EVALUATION

Three independent evaluation tracks exist, and all three are committed to [docs/benchmarks/](#) rather than quoted from memory.

6.1 Risk Engine bake-off

Four artifacts were trained and scored head to head. Because each model’s own held-out split differed in size (the shipped model held out 4,001 rows from a 20k population, the bake-off models 20,001 from 100k), scoring them on their own splits is not a leaderboard. We therefore generated a **fresh 20,000-patient population at seed 4242** — training used 1337, so every row is unseen by every artifact — and scored all four on identical data. No model was retrained.

Three findings, and the third is uncomfortable.

The shipped model leads where it must. It is best on discrimination, best calibrated by a factor of three, and the *only* candidate that holds conformal coverage at or above the 90% it promises. A confidence band that does not cover at its stated rate is worse than no band, so this gate is not negotiable — and it is the gate the three 100k models fail.

The 100k models are safer and less honest. XGBoost achieves a green-miss rate of 0.008 and an almost flat FNR spread of 0.028 across strata — genuinely better safety numbers. It buys them by being systematically over-confident: calibration slope 0.62 and ECE 0.073, nearly seven times the shipped model’s. It escalates more or less indiscriminately, which lowers misses and destroys the meaning of the number on the card.

No learned model reliably beats the hand-coded rule card. Comparing under-triage at four matched over-triage rates with bootstrap CIs, every candidate records exactly one significant win — at the loosest operating point, 0.50 — and ties or loses everywhere else. The `beats_handcoded_baseline` gate fails for all four on this fresh sample. We report this rather than omit it, because it is the strongest available argument for the architecture we chose: the deterministic rule card is the *floor* of the system, not a fallback we hope never to reach, and the model earns its place by watching trajectories over time — which the rule card structurally cannot do — rather than by beating it on a single snapshot.

6.2 The eight gates, side by side

Discrimination is only three of the eight gates. The other five ask whether the number is honest, whether the misses are equitably distributed, and whether the model earns its place at all (Fig. 3).

6.3 The safety metric, not accuracy

Accuracy is meaningless at 10% prevalence. The metric that governs is under-triage at a fixed over-triage rate: how many critical patients are left below Yellow when the department can tolerate a given false-alarm load.

6.4 Calibration, coverage and fairness

Fairness is defined as **equalised false-negative rate across strata** with bootstrap CIs — never a bare max–min gap, which at these positive counts would be mostly noise.

Calibration is strong and uniform: overall ECE 0.011 with a reliability slope of 0.954 and intercept -0.083 , and per-stratum ECE between 0.017 (neonate) and 0.033 (adolescent), reportable in all six. Conformal coverage is 0.910 overall with a mean set size of 1.08 — the guarantee is met while the set stays informative, which is the difficulty with conformal methods in practice.

6.5 Time to detection — the number the product is for

Everything above scores a *snapshot*. The claim this product actually makes is temporal, so we measured it directly. Every corpus patient carries a 37-step trajectory; we rebuilt the feature row at each 5-minute step through the same extractor the model was trained on, scored it with the shipped artifact, and read off two detection times from the same curve — the first 5-minute tick that crosses the threshold, and the first *scheduled* physical re-measurement at or after it on the band’s own cadence ([eval/time_to_detection.py](#)).

Sixteen of twenty patients cross the Yellow threshold during the wait. Across those, continuous re-scoring gets there a mean of **21.6 minutes** earlier than the scheduled check, median 17.5, maximum 55. Two patients would have lost a full cadence. At the Red threshold, which eight patients reach, the mean gain is 6.9 minutes — smaller, and honestly so: Red crossings tend to be dramatic enough that the shorter Yellow-band cadence catches them soon anyway. *The value is concentrated in the Yellow queue, which is exactly where the brief says people die.*

6.6 8/8 or 7/8 — and why the difference is the point

No artifact is deployable until all eight gates pass on the site’s own shadow-mode data. The shipped artifact clears **8/8** on its own committed split ($n=4,001$) and **7/8** on the fresh common split ($n=20,000$). Exactly one gate moves, and it is gate 8. On the smaller split the model wins at otr 0.50

Table 2. Risk Engine bake-off on a common held-out split (20,000 rows, prevalence 0.1028, generator seed 4242). Committed to [docs/benchmarks/risk_engine_bakeoff_table.md](#). The severity-oracle ceiling on this split is AUPRC 0.408 / AUROC 0.870; the prevalence floor is 0.103.

Model	Train n	AUPRC	AUROC	% of ora- cle	Green- miss	FNR spread	ECE	Conformal cov.
medipilot-gbdt-v0.2.0 (shipped)	20,000	0.1637	0.6689	40.1%	0.0948	0.2050	0.0110	0.9060
medipilot-hist-100k	100,000	0.1570	0.6489	38.5%	0.0375	0.1450	0.0430	0.8986
medipilot-hist-mono-100k	100,000	0.1566	0.6668	38.4%	0.0209	0.0795	0.0351	0.8961
medipilot-xgboost-100k	100,000	0.1549	0.6469	37.9%	0.0078	0.0284	0.0729	0.8974

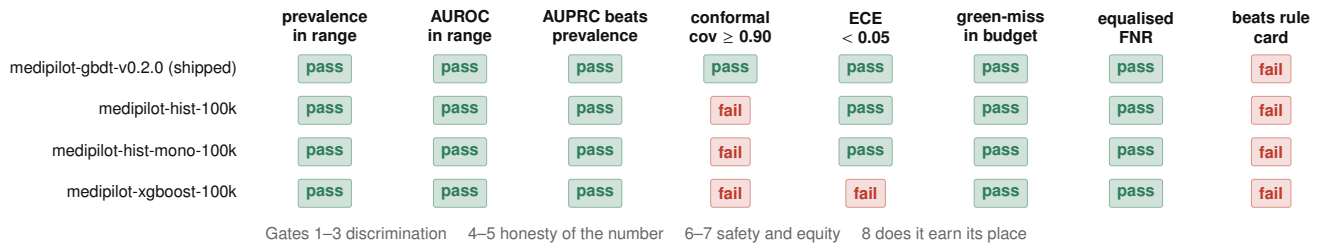


Figure 3. The eight go-live gates across the bake-off, on the common held-out split. No artifact is deployable until all eight pass *on the site's own shadow-mode data*. Two columns carry the story: gate 4 (conformal coverage) is where the three 100k models fail — a confidence band that does not cover at its stated rate is worse than no band — and gate 8 fails for every candidate, because none reliably beats the hand-coded rule card at tight operating points. The shipped model is the only one that clears the honesty gates, which is why it ships.

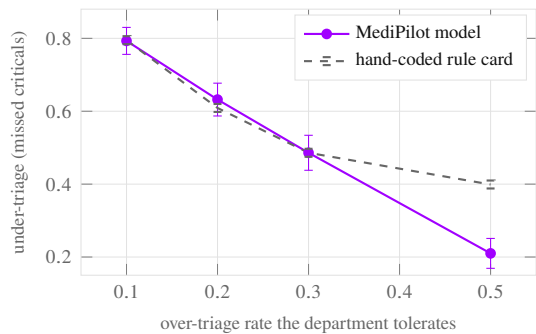


Figure 4. Under-triage at four matched over-triage rates, with bootstrap 95% CIs, on the shipped model's committed split. The curves are statistically indistinguishable until the department can tolerate a 50% over-triage load, where the model halves the misses (0.210 vs 0.399). This single separation is the honest extent of the learned model's measured advantage on snapshot scoring.

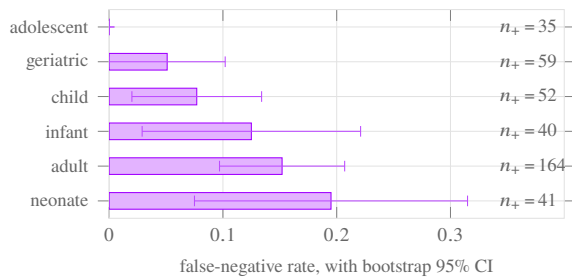


Figure 5. False-negative rate by age stratum for the shipped model, with bootstrap CIs and positive counts. The spread is 0.195. Neonates are the worst-served stratum on 41 positives — a real finding with genuinely wide error bars, and the single clearest argument for the paediatric data partnership in Phase 2 (§11).

and loses nowhere, because the intervals are too wide to separate; on the split five times larger they tighten until a loss at otr 0.20 becomes significant, and the gate flips. **We report the stricter result.** The full ledger, both splits side by side, is Table 3.

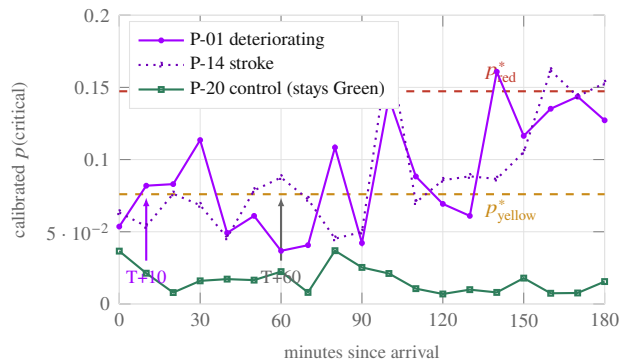


Figure 6. Calibrated risk over the full 180-minute wait, scored every five minutes with the shipped artifact. **P-01** crosses the Yellow threshold at **T+10**; the next *scheduled* physical re-measurement on the Green cadence would not arrive until **T+60** — a 50-minute head start. **P-20** is the control and never approaches the threshold, which is what makes the comparison meaningful. The curves are deliberately not smoothed: real vitals are noisy, and the escalation-only invariant is what makes that noise safe — a spurious high reading can only ratchet priority *up*, and a spurious low one cannot pull it down.

6.7 LLM structurer bake-off — ten candidates

The language model is confined to perception, and this evaluation is how that confinement is *measured* rather than asserted. Ten candidates were scored on a deliberately messy case set covering adversarial, ambiguous, negation, Hinglish, prompt-injection, obstetric, paediatric, stroke and reliability slices.

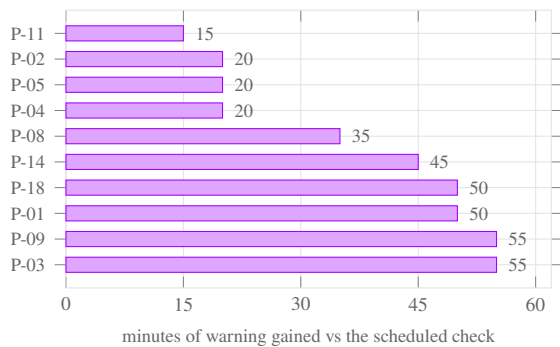


Figure 7. Head start per patient at the Yellow threshold: how many minutes earlier continuous five-minute re-scoring reaches the threshold than the next scheduled re-measurement would. Across the 16 of 20 corpus patients who cross at all, the mean gain is **21.6 minutes** and the median **17.5**. Ten patients are shown; the remaining six cross at T=0 and gain nothing, because they were already deteriorating on arrival — those are cases the initial assessment catches, and the system is not claiming credit for them.

Candidate	F1	RF miss	Schema fail	Lat. (s)
gpt-oss-120b (chosen)	0.962	0	0	7.59
gpt-oss-20b (chosen, fast)	0.816	4	4	8.89
Qwen2.5-7B (local)	0.653	7	0	5.46
Qwen2.5-3B (local)	0.650	8	0	6.42
Phi-4-mini (local)	0.608	3	0	5.65
Qwen2.5-1.5B (local)	0.528	11	0	4.04
Rule baseline (no LLM)	0.510	12	0	0.00
llama-3.3-70b	0.325	18	40	0.15
llama-3.1-8b	0.325	18	40	0.10
kimi-k2-instruct	0.325	18	40	0.08

Two results matter commercially. First, **the three largest instruction-tuned models are the worst performers here** — 40 schema failures each, meaning they would not emit the constrained form at all. Capability and constrained-decoding compliance are different properties, and buying the biggest model would have been the wrong decision. Second, **the rule baseline misses 12 red flags and the chosen model misses zero**, which is exactly why the deterministic structurizer is the floor and not the default — and why losing the network degrades the system rather than stopping it.

6.8 ASR bake-off — twelve backends

Transcription was selected on measured word error rate, not on a datasheet. `whisper-large-v3-turbo` through the hosted path and `whisper-local:turbo` both reach WER 0.000 on the reference utterances, with hosted latency 0.30s against 1.21s locally and a 4.94GB VRAM footprint for the local path. Nine other backends — including `faster-whisper:large-v3` at both `int8_float16` and `float16`, which score WER 0.583 — were rejected on measurement. **Zero silence hallucinations were recorded across every backend**, which is the result the pre-flight RMS silence gate exists to guarantee.

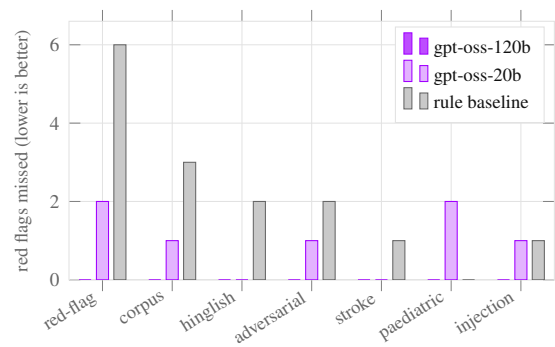


Figure 8. Red flags missed per evaluation slice — the only structurizer metric with a body count. `gpt-oss-120b` misses none anywhere, including the code-mixed Hinglish and prompt-injection slices. The rule baseline misses six on the red-flag slice alone and is the only candidate that fails on stroke. Note the *paediatric* column: `gpt-oss-20b` misses two where the rule baseline misses none, which is precisely why the 20b model is confined to option matching and classification while the 120b does observation extraction.

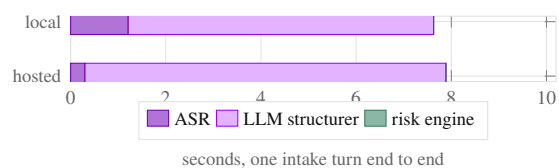


Figure 9. Where the time actually goes in one intake turn. The **risk engine is 3 ms** — a measured p50 over 300 calls, with p95 4.3 ms and p99 8.0 ms — and is invisible on this scale. Effectively all latency is perception, and the structurizer dominates it. This is an architecturally useful result: the expensive component is the one that may only *report*, so it can be degraded, cached or dropped to the offline tier without touching the component that decides.

6.9 The invariant suite

282 tests run on every change, of which 278 pass; the four failures are a packaging artefact and a stale assertion, both documented in the implementation README. The suite is not a unit-test formality — it is where the clinical constraints are enforced: 15 invariant tests, 10 on age stratification, 10 replaying the corpus acceptance cases, 8 on surge control, 7 on the band engine, 6 on the audit log and 6 on reliability weighting.

7 TARGET USERS AND ADOPTION

7.1 Five users, three of whom are not clinicians

User	What they get, and what they can do
Triage nurse	The primary user. One card, two supporting factors, one opposing factor, a confidence band. Accept or override in one touch. Gains a continuous monitoring layer she cannot physically provide
Patient	Speaks Hindi or English to a kiosk, or taps. Never told a colour out loud. Gets a real wait estimate and is told early if it slips
Family / attendant	Can perform and close a <i>Green</i> recheck only; doing so raises a flag for staff. Recognises that families already participate in care in Indian EDs, without granting clinical authority
Counter / office staff	Registration and token issue. Sees queue position, never risk scores
Clinical governance	The audit ledger, override patterns, drift alarms, and sign-off on every YAML change

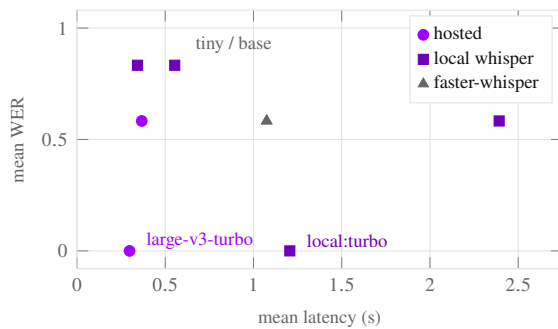


Figure 10. ASR selection on measurement rather than on a datasheet. Two backends reach WER 0.000; the hosted turbo path is 4× faster than the local one, and the local path costs 4.94 GB of VRAM. Everything at WER 0.583 and above was rejected — including both faster-whisper:large-v3 quantisations, which a specification sheet would have ranked highly. faster-whisper:small/int8 is off-scale at WER 3.750. **Caveat:** only two reference utterances, so this separates the field but does not settle it (§5).

7.2 Adoption — designed against the failure modes of clinical AI

Fatigued staff work around tools that add work or insult their judgement. Six choices follow directly:

- **Zero workflow change to start.** The wedge module never assigns an initial acuity; it watches a queue the nurse already ordered. Nothing is taken away.
- **The opposing factor.** Every card names the strongest evidence *against* its own recommendation. This forces engagement with reasoning rather than with a number, and it is the standard defence against automation bias.
- **Override is one touch and is never punished.** It is a first-class route in the API, not an escape hatch.
- **Abstention is visible.** A system that says "I don't know" on the ~8% it genuinely cannot judge earns trust on the remainder.
- **No virtual-nurse persona.** No avatar, no manufactured clinical authority.
- **Shadow mode first.** Staff see the system's judgement compared against their own for weeks before it is allowed to surface anything.

8 DATA PROTECTION AND REGULATORY POSITION

8.1 Stated jurisdiction: India

DPDP Act 2023, the ABDM consent framework, ICMR's 2023 AI ethics guidance, and the CDSCO draft guidance on AI/ML medical device software including algorithm change protocols.

The gap we name rather than paper over. ABDM consent for care delivery does *not* by itself satisfy DPDP Act 2023 for using the same data as model input. These are different purposes and require separate, revocable consent. Any vendor claiming ABDM integration equals DPDP compliance for training is wrong, and this is the single largest regulatory item on our roadmap.

8.2 How the data is actually protected

On-premise by default. With no API credential configured the system runs entirely inside the hospital: local ASR, offline phrase matching, the deterministic structurizer, and no outbound request on any path. Privacy is a deployment mode, not a policy document.

Only model updates travel. Cross-site learning is federated with secure aggregation — weights move, patient data does not. Per-site calibration stays local.

The audit ledger. Every autonomous decision and every override appends a record with 17 substantive fields plus two hash-chain fields. It is append-only and SHA-256 hash-chained, so a retroactive edit is mathematically detectable. Consent withdrawal is recorded as a *new event*, never as a deletion, which is what preserves the integrity of the chain while honouring the withdrawal.

What a clinician override legally records. Named clinician and role, system band and clinician band, direction, reason code and free text, the calibrated score and confidence at that moment, the factors as displayed on the card, an inputs hash, model and calibration versions, and the consent state. `outcome_ref` is back-filled when known — and that back-fill is the training signal.

9 BUSINESS CASE AND IMPACT

9.1 Unit economics

The figures below are an **illustrative model under stated assumptions**, not measured revenue. No pilot has been run and no price has been agreed.

Assume the brief's range of 100–500 visits per day. The architecture is deliberately cheap because the required path has no GPU and no per-token cost: the risk model is gradient-boosted trees at millisecond latency on commodity CPU, and the language model is optional. The white paper's design target is **under Rs. 5 per encounter** at the edge tier.

Site	Visits/day	Enc./yr	At Rs. 5
District hospital	100	36,500	Rs. 1.8 L/yr
Mid-size urban ED	250	91,250	Rs. 4.6 L/yr
Tertiary trauma centre	500	182,500	Rs. 9.1 L/yr

9.2 Where the value lands

The economic case does not rest on staff replacement — headcount is unchanged. It rests on four effects, ordered by how directly they are measurable:

1. **Deterioration caught during the wait.** The single event the system exists to prevent. Each avoided in-queue deterioration avoids an ICU escalation.
2. **Fewer left-without-being-seen.** Green patients receive a real wait estimate and are told when it slips. LWBS is already tracked by most EDs, so this is measurable from day one without new instrumentation.
3. **Documented defensibility.** A hash-chained audit trail of every recommendation and every override is a liability-management asset independent of clinical outcome.
4. **Compounding calibration.** Each site improves the model for every other site through federated rounds — an advantage structurally unavailable to a single-site or late entrant.

9.3 Why this is defensible

A well-funded competitor can rebuild the technical stack in a year. It cannot rebuild years of site-specific calibration and federated override history across dozens of Indian EDs. The

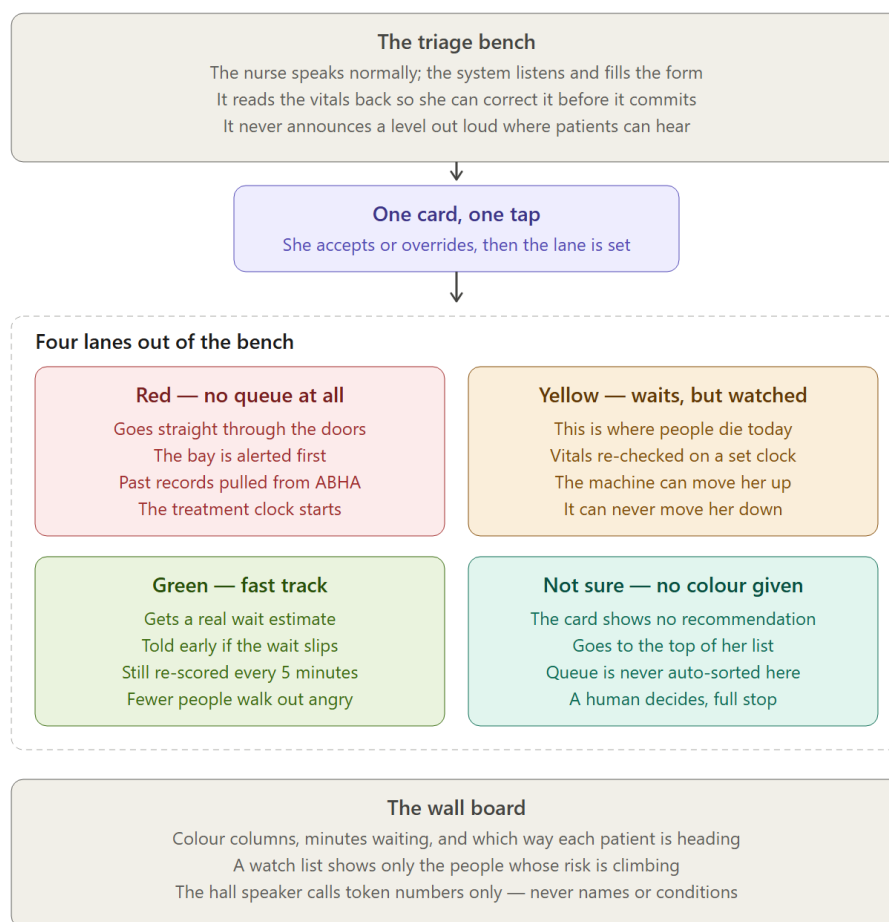


Figure 11. What each user actually experiences. The abstention lane is a first-class outcome rather than an error state: no colour is offered, the queue is never auto-sorted, and a human decides.

moat is the data relationship, not the model.

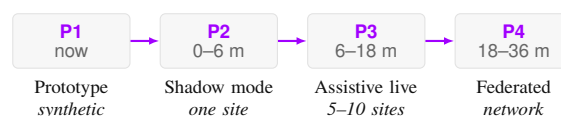
10 SCALABILITY ACROSS HOSPITALS

The brief asks how one assistant flexes across hospitals of very different size, specialty mix and technical maturity. The answer is that **the tiers are independently adoptable** (Fig. 12).

Maturity	Deploys	What they get
No IT department	Bench box only	Offline edge inference, SQLite, browser PWA. Works with the network cut
Has a network	+ hospital tier	Board and phone on the same build, 5-minute event ticks
Has an EHR	+ FHIR bridge	The card appears inside the existing workflow via CDS Hooks
Multi-site group	+ regional cloud	Federated rounds, drift watchdogs, self-hosted structurer

The honest engineering constraint. Today both back-end applications hold state in process — the orchestrator bootstraps a *World* in its lifespan hook and runs a tick loop. Horizontal replication therefore requires externalising that state to Postgres and Redis *first*; attempting workers before that produces a system that looks scaled and silently disagrees with itself. This is the first engineering item in Phase 2, not an afterthought.

11 PHASED ROADMAP



Each phase has an exit gate that is a measurement, not a date.

Phase	Work	Exit gate
P1	Working prototype on synthetic data. <i>Complete.</i> 7 surfaces, 20-patient corpus, 4-model bake-off, 282-test suite	Every mandated case demonstrable end to end — <i>met</i>
P2	Externalise state (Postgres + Redis). Partner site, DPDP/ABDM consent instrument, ethics approval. Shadow mode: model scores silently, nothing surfaced. Paediatric data partnership	8/8 go-live gates on the <i>site's own</i> outcome data; neonate FNR CI narrowed below the adult point estimate

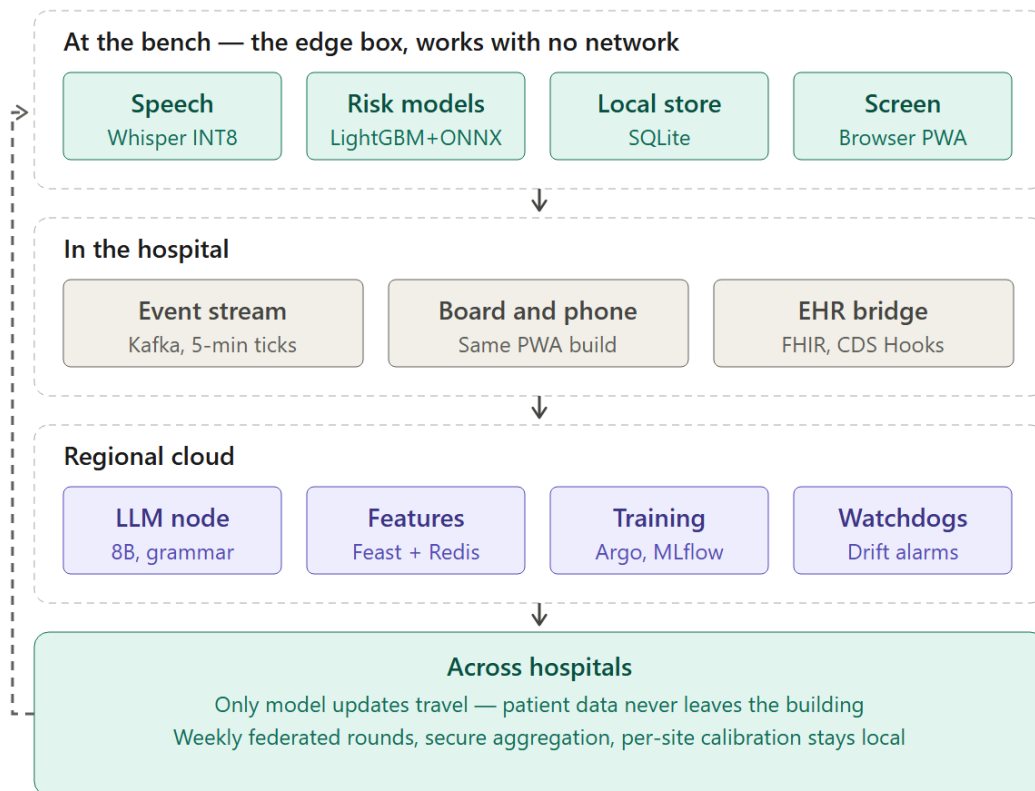


Figure 12. Three deployment tiers. A hospital adopts only the tiers it can support: the bench box alone is a complete product. The bottom row is the invariant — only model updates travel between sites, and per-site calibration stays local.

Phase	Work	Exit gate (<i>cont.</i>)	Risk	Mitigation (<i>cont.</i>)
P3	Assistive deployment: recommendations surfaced, nurse decides. FHIR/CDS Hooks bridge. ONNX edge box. CDSCO SaMD submission with a predetermined change control plan	Stepped-wedge cluster randomisation showing reduced in-queue deterioration — <i>not</i> a pre/post comparison	Label leakage	The failure that broke the Epic Sepsis Model. Mitigated mechanically: leakage classes in the feature registry, fail-closed on unregistered fields, and a CI canary test
P4	Federated rounds with secure aggregation across the network. Drift watchdogs per site. Self-hosted structurer	Cross-site model improvement demonstrated with zero patient-data egress	Automation bias	Nurses defer to the screen. Mitigated by the mandatory opposing factor, visible abstention and override telemetry under clinical governance
			LLM emits a judgement	Structurally impossible rather than discouraged: no production rule for acuity exists in the output grammar; tier-2 output vocabulary is closed on both sides of the wire
			Pulse-oximetry bias	Occult hypoxaemia is 3× more common in darker-skinned patients. A normal SpO ₂ carries <i>no</i> de-escalation authority, and pallor calibration is Fitzpatrick-stratified
			Network / power loss	Offline-first edge inference, UPS operation, store-and-forward sync. The bench box works with the network cut
			Regulatory delay	The wedge module never assigns an initial acuity — the lowest-exposure entry point. CDSCO submission carries a predetermined change control plan, so retraining does not require resubmission
			Surge misuse	Three actions raise <code>SurgeViolation</code> rather than being discouraged: raising <i>R</i> , guessing an abstention, de-escalating for capacity

12 KEY RISKS AND MITIGATIONS

Risk	Mitigation — and where it is enforced
Synthetic-to-real gap	The largest risk in this proposal. No absolute metric transfers. Mitigated only by shadow mode re-fitting calibration on site data before anything surfaces; the go-live gates are evaluated on the site's data, never ours
Model no better than rules	<i>Measured, not hypothetical</i> (§6). Mitigated architecturally: the rule card is the floor, so a model that adds nothing subtracts nothing. The model's distinct value is temporal, and that is what P3 must demonstrate
Neonatal under-triage	Worst stratum at FNR 0.195. Mitigated by the paediatric data partnership gate in P2 and by per-stratum thresholds; until then neonates carry the most conservative cut

APPENDIX — THE GATE LEDGER, READ FROM THE FILES

Nothing in §6 is asserted. Every value below is read from a committed artifact, and each gate is shown *twice*: on the shipped model's own held-out split ($n=4,001$, seed 1337) and on the fresh common split ($n=20,000$, seed 4242) built

for the bake-off, on which no artifact had ever been scored.

#	Gate	Rule	Own split	Fresh split
1	Prevalence in range	0.05–0.20	0.098 pass	0.103 pass
2	AUROC in range	0.60–0.95	0.689 pass	0.669 pass
3	AUPRC beats prevalence	> 1.5× prev.	0.174 pass	0.164 pass
4	Conformal coverage	≥ 0.90	0.910 pass	0.906 pass
5	ECE where reportable	< 0.05	0.011 pass	0.011 pass
6	Green-miss in budget	≤ 0.15	0.115 pass	0.095 pass
7	Equalised FNR spread	≤ 0.25	0.195 pass	0.205 pass
8	Beats the rule card	≥ 1 win, 0 losses	win @0.50 no loss pass	win @0.50 loss @0.20 FAIL
Verdict			8/8	7/8

Table 3. The eight go-live gates for `medipilot-gbdt-v0.2.0`, evaluated on both splits. Source: `model/artifacts/medipilot-gbdt-v0.2.0/metrics.json` and `docs/benchmarks/risk_engine_bakeoff_results.json`.

Two entries deserve a note rather than a footnote. **Gate 6** admits a 0.15 green-miss rate while the threshold solver *targets* 0.10 on calibration data; the shipped model lands at 0.103 on the data it was solved on and 0.115 held out. That gap is ordinary generalisation slack, inside the gate and outside the solver’s aim, and it is reported rather than tuned away. **Gate 8** is the honest failure of this proposal, and §6 treats it as the central architectural argument, not a footnote: a rule card that a learned model cannot reliably beat is the correct floor to build on.

Why a fresh split, and why one model is missing

The common split was generated at seed 4242, not the training seed 1337. Replaying 1337 would have handed each model back its own training rows; worse, the artifacts held out different sample sizes — 4,001 for the shipped model against 20,001 for each 100k candidate — so their own splits are not comparable to *each other*. Scoring five models on five different samples is not a leaderboard. Every row of the fresh split is unseen by every artifact, which makes it the stricter comparison as well as the fairer one.

One candidate is absent from Fig. 3 and from the ledger. `medipilot-last-obs-100k` is a sequence baseline that unpickles through `model/sequence_models.py`, which imports torch at module scope; torch is installed in this environment but its DLL fails to initialise, so no row could be produced. It is recorded as *excluded* in the results file rather than quietly dropped.

Per-stratum fairness, in full

Gate 7 compresses six numbers into a spread. The six, at the shipped per-stratum Yellow cuts:

Stratum	FNR	n_+	95% CI
Adolescent	0.000	35	0.000–0.000
Geriatric	0.051	59	0.000–0.121
Child	0.077	52	0.020–0.160
Infant	0.125	40	0.029–0.237
Adult	0.152	164	0.097–0.210
Neonate	0.195	41	0.075–0.325
Spread 0.195		gate ≤ 0.25 pass	

The neonate interval (0.075–0.325) *overlaps* the adult one (0.097–0.210) on 41 positives. The worst stratum is therefore not yet statistically distinguishable from the best-powered one — which is precisely why the Phase 2 exit gate is written as *narrowing that interval*, not as hitting a point estimate. Claiming a fairness gap here would overstate the data exactly as confidently as denying one.

Where every number in this document comes from

Claim	Committed artifact
Risk Engine bake-off, 4 models × 8 gates	<code>docs/benchmarks/risk_engine_bakeoff_results.json</code>
Shipped metrics, gate verdicts, per-stratum FNR	<code>backend/model/artifacts/medipilot-gbdt-v0.2.0/metrics.json</code>
Head start, $p(r)$ curves, serving latency	<code>docs/benchmarks/time_to_detection.json</code> , generated by <code>backend/eval/time_to_detection.py</code>
LLM structurizer bake-off, 10 candidates	<code>docs/benchmarks/structurer_bakeoff_results.json</code>
ASR bake-off, 12 back-ends	<code>docs/benchmarks/asr_bakeoff_results.json</code>
The 20-patient corpus	<code>backend/data/corpus_20.json</code>
Band cadence, thresholds, red flags	<code>backend/config/band_cadence.yaml</code> , <code>red_flags.yaml</code>
Invariant suite (282 tests)	<code>backend/tests/</code>

MediPilot • “Nobody waits unseen.”

WHERE THE BRIEF IS ANSWERED

Brief pointer	Section
Data strategy; mixed data availability	§2, §5
Decision model; confidence indicator	§2, §6
Workflow design; adoption and change mgmt	§2, §7
Safety-first defaults; waiting-queue monitoring	§2
Age-varying thresholds	§4
Patient data protection; jurisdiction	§8
Scalability across hospitals	§10
15–20 records; ambiguous, paediatric, geriatric, no history	§3, Tab. 1
3× surge behaviour	§2, §3
Clinician override and what is logged	§3, §8

Provenance. Every quantitative claim traces to a committed artifact under `docs/benchmarks/` or `backend/`. Implementation detail is in the companion `implementation_readme.pdf`.